

Narula Institute of Technology

NAAC 'A' Accredited | NIRF Ranked College

Database Management Systems – Internal Assessment Question Paper

Time: 1 Hour

Maximum Marks: 25

Instructions: Answer all questions. Each question carries 5 marks. Draw neat diagrams wherever required.

Q.	Question	CO	BT	Marks
1	Compare RDBMS and NoSQL databases in terms of scalability and schema flexibility.	CO1	BT2	5
2	Design a complete E-R diagram for a Hospital Management System and convert it into relational tables.	CO2	BT6	5
3	Analyze how the Three Schema Architecture maintains data independence in a dynamic environment.	CO1	BT4	5
4	Consider the following relations and solve the queries using Relational Algebra: Student (SID, Name, Dept, Age) Course (CID, CName, Credits) Enroll (SID, CID, Grade) (a) Find the names of students enrolled in the course with CID = 'DB101'. (b) Retrieve the names of students who obtained grade 'A'. (c) Find the names of students along with the names of courses they are enrolled in.	CO3	BT3	5
5	What is cardinality in DBMS? Explain its types.	CO2	BT1	5

CO (Course Outcome): CO1 – DBMS concepts and architecture; CO2 – Data modelling (E-R model); CO3 – Relational model and algebra.

BT (Bloom's Taxonomy): BT1 Remember, BT2 Understand, BT3 Apply, BT4 Analyze, BT5 Evaluate, BT6 Create.

***** End of Question Paper *****